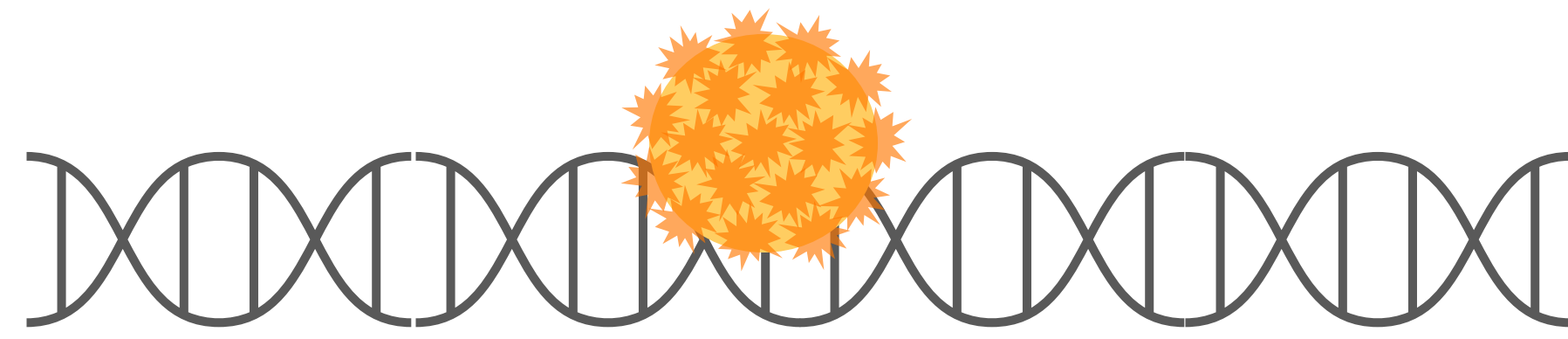


INTRODUCTION

Human Papillomavirus (HPV) is a common virus that can be transmitted through skin-to-skin contact. When high risk HPV infects the human cell, its DNA can integrate into the human genome.

Viral Integration Sites (VISs) are specific locations in DNA where viruses can become incorporated within the human genome. These integrations are the cause of about 15% of human cancers.¹



Goal: To develop a deep learning model to predict whether DNA sequences represent oncogenic VISs, thereby aiding in the identification of potential cancer-driving events at the molecular level.

DATA COLLECTION & PREPROCESSING

Positives: The HPV VIS data had 70,165 sites with positions and insertion locations, which were used to flank each sequence by 500bp upstream and downstream, resulting in 1000bp sequences. Then, “*BEDTools getfasta*” was used to extract sequences from human genome reference.

Negatives: Negative sequences were generated using “*BEDTools*” to identify regions not covered.

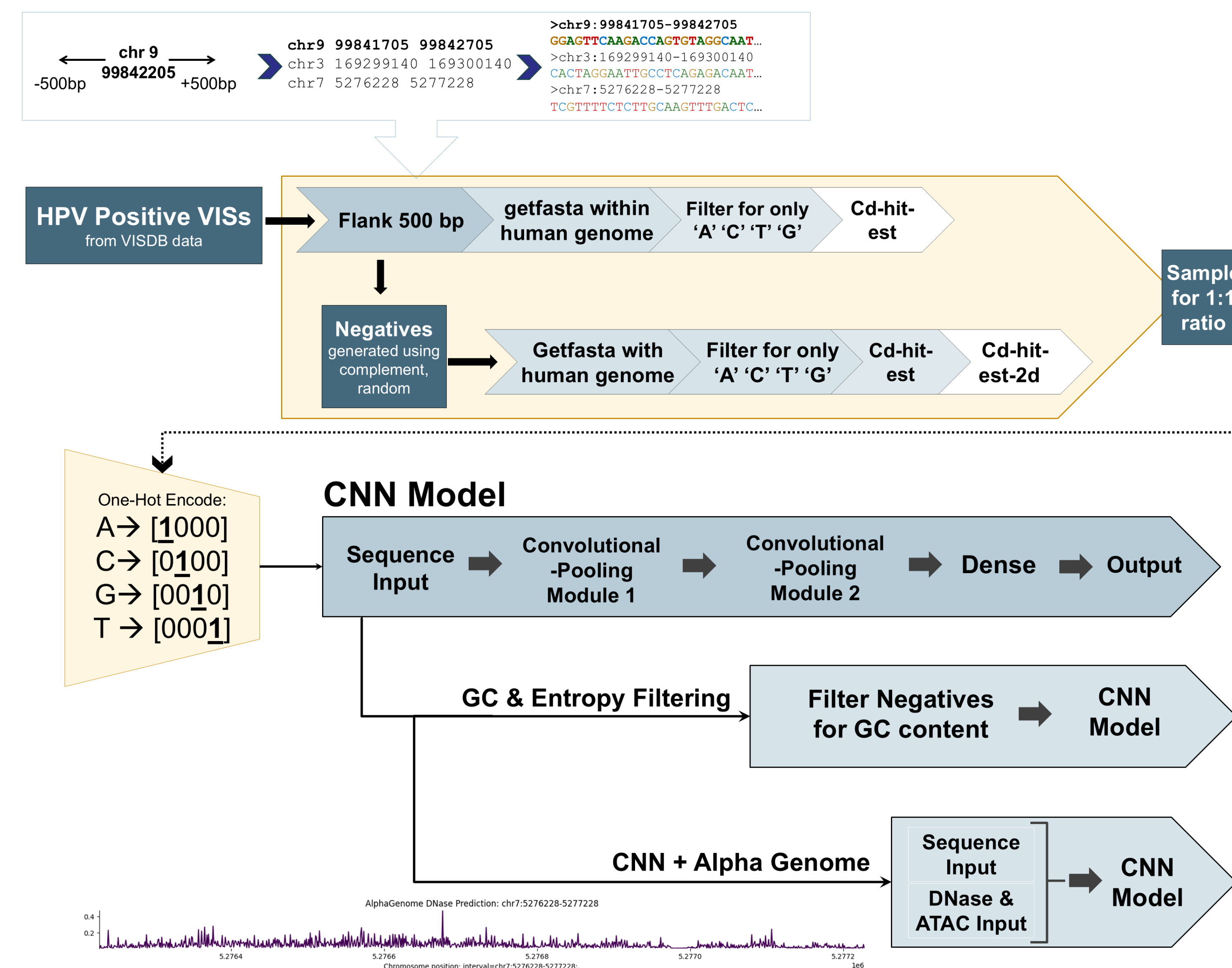
Filtering: Sequences were filtered to only contain ‘ACGT.’ To remove redundant sequences within the dataset, cd-hit-est and cd-hit-est-2d were used with a 90% similarity threshold, resulting in 61,082 positive sequences, and negatives were then randomly sampled to obtain the same ration as positives.

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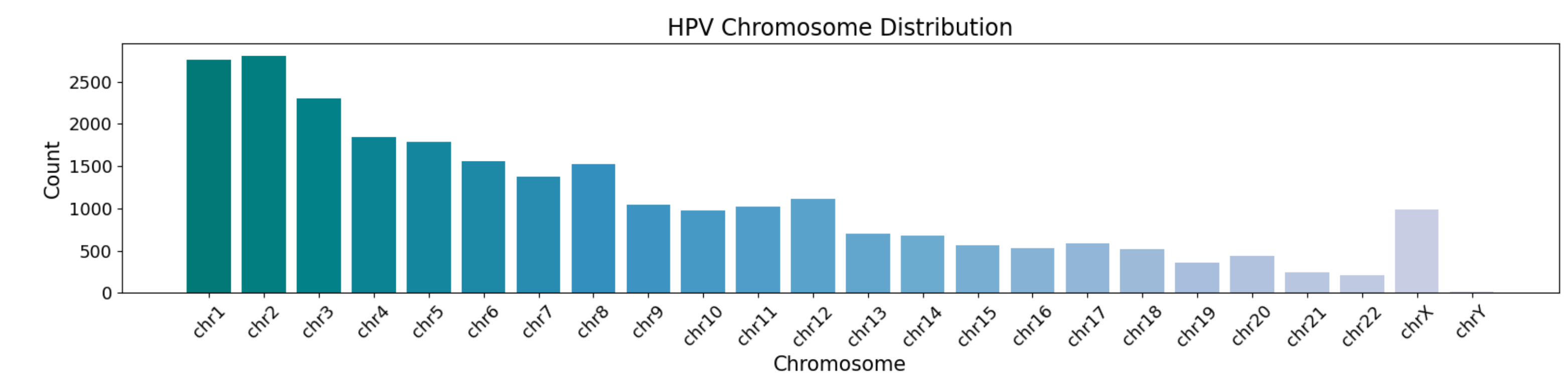
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METHODS

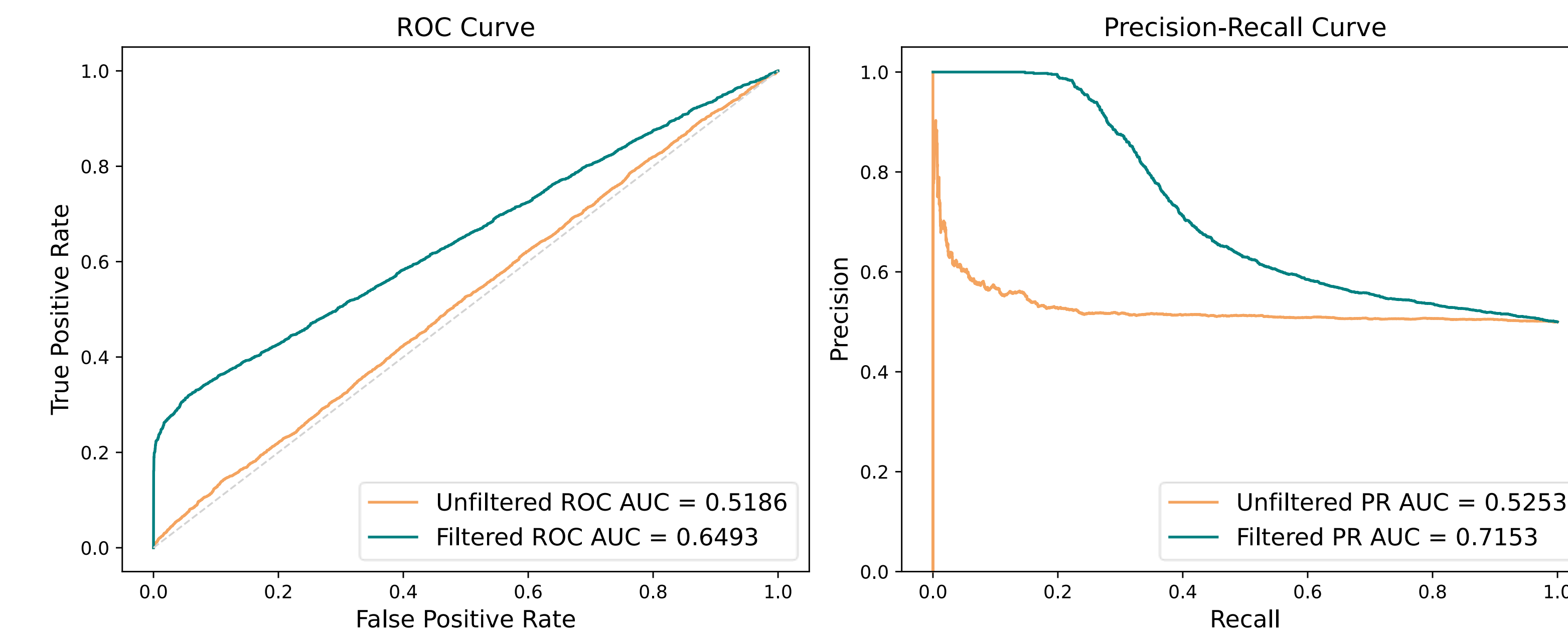
- Using Convolutional Neural Network (CNN) architecture from other VIS CNNs as a guide ^{1 2}, a similar model was created with an input layer, Conv1D/batch normalization/pooling blocks, with average pooling and dense layers.
- The effect of GC content is commonly corrected for.⁴ To address this, 15% of negative samples with the highest and lowest GC content were excluded. With the filtered data, the ROC improved from 0.52 to 0.65, and PR improved from 0.52 to 0.72.
- Separately, AlphaGenome ³ was used to obtain DNase and ATAC from the sequences to improve predictions. Adding these as additional input improved results further, with a PR of 0.65 and ROC of 0.76.



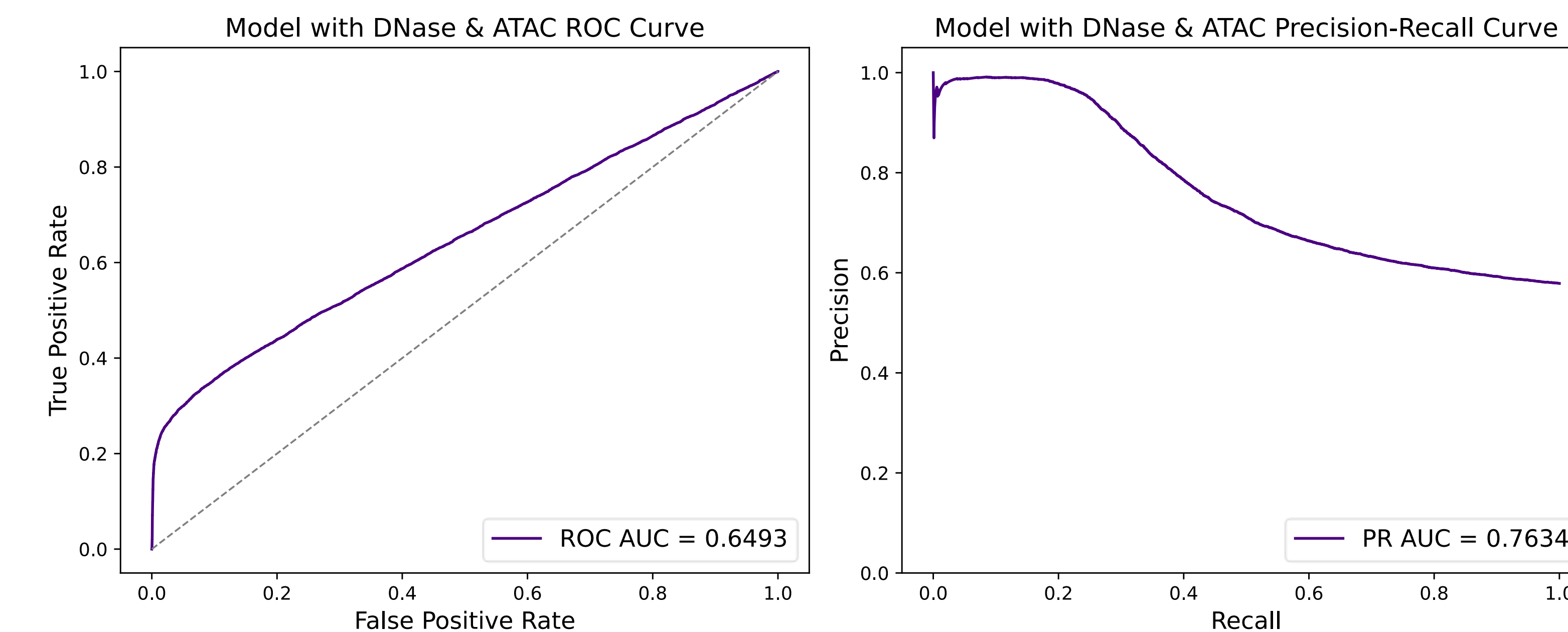
RESULTS



A. Frequency of HPV VISs by chromosomes.



B. Receiver Operating Characteristic Curves and Precision-Recall Curves for the CNN model for both the original data and the data with negatives filtered for GC content.



C. Receiver Operating Characteristic Curve and Precision-Recall Curve for the CNN model including both sequence input and DNase and ATAC inputs from AlphaGenome.

CONCLUSIONS

- The CNN models successfully fit HPV VIS data, but they currently show limited generalization beyond the training set, indicating the need for further optimization.
- Filtering the negative samples for GC content increased performance.
- Adding AlphaGenome features led to an improvement in model performance.

Limitations

- Filtering negatives for GC content could introduce bias that limits generalizability.
- Currently, the models reveal signs of overfitting, with higher training performance than validation performance.

Next Steps

- With additional resources, more complex models and larger datasets could further enhance performance.
- Additionally, the performance of Transformer models rather than CNNs could be tested with more time and resources.

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